

Systemic Risk? The System is at Risk

Why Quantum Computing Demands a New Model for Financial Stability



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Quantum is not a footnote in the story of financial innovation — it's a t shift in the architecture of systemic risk. For decades, regulators and c banks have relied on a relatively stable playbook: identify systemically important financial institutions (SIFIs), supervise them more intensely, manage contagion risk through capital buffers, stress tests, and recovery plans. That model worked — more or less — in a world where systemic was a function of size, *interconnectedness*, and *complexity*.

Quantum computing doesn't fit that model. In fact, it blows it up.

Once cryptographically relevant quantum computers (CRQCs) become reality, they will undermine the cryptographic foundations that every f institution depends on — regardless of its size, business model, or regulatory designation. What we are facing is not an institution-specific vulnerability, but a *universal* one. And unlike traditional risk, quantum risk is not gradual, idiosyncratic, and not containable.

This is what makes it systemic in the most literal sense: the entire system is at risk.

Disclaimer: Current Quantum Limitations

While the risks outlined here are real, it is important to recognize the limitations of quantum computing. Today's quantum devices remain no

error-prone, and resource-constrained. Quantum error correction (QEC) — the process required to stabilize qubits and scale toward cryptographically relevant quantum computers — is still in its infancy. Current experiments show progress, but the overhead remains enormous: achieving a single logical qubit may require thousands of physical qubits, and fault-tolerant architectures are not yet practically viable.

This means that quantum attacks on widely used cryptographic standards are **not possible with existing hardware**. The concern lies in the **closing preparation window**: migration to post-quantum cryptography will take a decade or more, adversaries are already harvesting encrypted data now, and the pace of breakthroughs is uncertain. The prudent approach is to plan for disruption early, rather than wait for certainty.

The Quantum Threat Landscape: No Longer Theoretical

Until recently, quantum computing felt like a “next-decade” concern — important to track, but not urgent to act on. That timeline is collapsing.

In 2024, Google’s *Willow* experiment marked a significant leap in quantum error correction, narrowing the gap between experimental quantum systems and cryptographically relevant ones. Shor’s algorithm — which allows a quantum computer to factor large numbers exponentially faster than classical computers — remains the biggest threat to RSA-2048, the backbone of financial encryption.

The expert community is no longer debating if CRQCs will arrive, but when. Estimates now range from the early 2030s to the late 2050s. But those bounds may offer false comfort. As recent research and private-sector

advances in error correction and algorithm design show, the field is accelerating faster than conservative models assumed.

Meanwhile, the National Institute of Standards and Technology (NIST) finalized its first three post-quantum cryptography (PQC) standards including ML-KEM for key establishment and ML-DSA for digital signatures. These are the building blocks of a quantum-safe future. But the vast majority of organizations are nowhere near implementation.

A 2025 ISACA survey found that:

- 62% of cybersecurity professionals believe quantum will break current encryption;
- Only 5% consider it a high near-term priority;
- And only 5% of organizations have defined a quantum strategy.

That gap between awareness and action is not a technical oversight — governance failure in the making.

The Hidden Front: "Harvest Now, Decrypt Later"

Perhaps the most urgent quantum-era threat is one that doesn't even require a working quantum computer today: the *harvest now, decrypt later* (HNDL) strategy.

Adversaries — state and non-state — are already intercepting encrypted financial data, storing it indefinitely, and waiting for the day they can crack it using quantum tools. That includes:

- Client communications
- Transaction records
- Proprietary algorithms
- Trading histories
- Interbank transfers

In other words, the data you think is safe today may already be compromised — just not yet decrypted.

For the financial sector, this is particularly dangerous because many categories of data have *long confidentiality lifespans*. A customer's identity may need to remain private for decades. A pension plan's trading strategy could still be sensitive in 2040. If that data is harvested today and decrypted in 2032, the damage will be real, retroactive, and irreversible.

This is where quantum risk becomes temporal: we are living in a vulnerable window that traditional risk frameworks simply don't account for.

Why Traditional Systemic Risk Models Break Down

A. The SIFI-Centric Model

After the 2008 crisis, regulators like the U.S. FSOC and the Basel Committee built systemic risk frameworks around a few core assumptions:

- **Size matters:** bigger balance sheets = bigger systemic impact.
- **Interconnectedness matters:** contagion spreads through networks.
- **Graduated oversight works:** tailor supervision based on institution

importance.

- **Failure is containable:** resolution and recovery are possible.

These principles underpin tools like:

- Capital buffers and G-SIB surcharges
- Stress tests
- Recovery and resolution planning
- Macroprudential tools

But these tools are institution-specific. They assume systemic risk is *differentiated* — that some firms are more critical than others.

B. Quantum Creates Flat Risk

Quantum computing breaks that assumption. It creates what some researchers call a “**flat risk profile.**” Why?

- Every institution relies on the same vulnerable cryptographic primitives: RSA, ECC, Diffie-Hellman.
- All are exposed *simultaneously* on “Q-Day.”
- Vulnerability is a function of technology, not balance sheet size.
- Contagion is not transmitted through networks — it occurs *systemically*.

That means:

- A small credit union is just as vulnerable as a major bank like JPMo
- Historical loss data becomes useless for predicting exposure.

- The logic of proportionality in regulation — more oversight for big firms — collapses.

As the Bank for International Settlements bluntly stated in 2025, “*trust financial system is fundamentally tied to the trust provided by cryptography*”. When that cryptographic trust is simultaneously undermined, the traditional model of systemic risk becomes obsolete.

The Crypto-Financial Frontier: Bitcoin, Ethereum, and Beyond

The quantum threat doesn't stop at traditional banks. The entire digital financial ecosystem is built on cryptography.

Bitcoin and Ethereum use elliptic curve digital signatures (ECDSA), which are highly efficient — and quantum-vulnerable. Roughly 31% of Bitcoin's total supply — over 6 million BTC — is already in addresses where the private keys are publicly exposed. Once quantum computers can break ECDSA, those assets could be stolen instantly.

This isn't just a crypto problem. Many institutions — hedge funds, payment companies, exchanges, custodians — hold digital assets. A successful quantum attack on a major blockchain would trigger:

- Mass theft of assets
- Loss of institutional confidence
- Liquidity crunches across markets
- Systemic losses for financial institutions with crypto exposure

Unlike traditional bank failures, which unfold over days or weeks, quantum attacks on crypto would be *instantaneous* and *irreversible*.

The Regulatory Response: Fragmented and Incomplete

A. Central Bank Awareness Is Growing

In July 2025, the BIS published its first quantum-readiness roadmap — an unprecedented move. It called the quantum threat “an imminent risk to financial stability requiring urgent, coordinated action.”

Some jurisdictions are responding:

- **United States:** Quantum-safe deadline for federal systems by 2035
- **Canada:** High-priority systems must migrate by 2031; all by 2035.
- **European Union:** ENISA and 18 member states issued joint calls for immediate action.

But this is a patchwork. Timelines differ. Standards differ. Regulatory complexity varies. There is no global mechanism yet for coordinated migration.

B. From Capital Adequacy to Crypto-Agility

A more fundamental shift is underway: the emergence of **crypto-agility** regulatory concept.

Crypto-agility means building systems that can:

- Rapidly switch to new algorithms
- Manage algorithm diversity

- Stay secure in multi-vendor, multi-jurisdictional ecosystems

This is not the same as capital adequacy or liquidity buffers. It's an operational resilience requirement — one that depends on technology architecture, vendor coordination, and long-term migration planning.

But adoption is slow. The Cross Market Operational Resilience Group (CMORG) recommends basic steps like cryptographic inventories and migration plans, yet few institutions have completed them. Post-quantum migration for large firms could take 10–15 years and cost hundreds of billions globally — rivaling the Y2K remediation effort.

Redefining Systemic Importance

If quantum risk is universal, how should regulators rethink “systemic importance”?

One option is to shift from size-based metrics to **resilience-based metrics** such as:

- **Cryptographic inventory completeness**
- **Migration progress toward PQC**
- **Crypto-agility maturity**
- **Vendor quantum readiness**
- **Cross-border interoperability**

Under this model, a small institution that is fully quantum-ready could be deemed *less systemically risky* than a global bank still running on legacy

cryptography.

The network structure still matters — but differently. During the transition, quantum-safe institutions will need to interact with quantum-vulnerable ones. That creates a new kind of operational risk: secure nodes become *islands* in a vulnerable sea. Payment systems, clearinghouses, and custody may acquire new systemic importance not by size, but by their role in *maintaining interoperability*.

Policy Implications and Recommendations

1. Build a Universal Quantum Risk Assessment (UQRA)

Regulators must treat *all* institutions as potential systemic flashpoints in a quantum context. That means:

- Moving beyond SIFI logic
- Supervising cryptographic readiness, not just capital ratios
- Creating crypto-agility requirements
- Requiring migration roadmaps and inventories

2. Redesign Stress Testing

Traditional stress tests model recessions, rate hikes, or liquidity shocks. Quantum-era stress tests must model:

- Sudden “Q-Day” scenarios
- Gradual quantum capability emergence
- Retroactive HNDL breaches

- PQC delays and standard fragmentation

These tests must evaluate infrastructure, not just solvency.

3. Coordinate Globally

Quantum risk does not respect borders. Coordination is essential to:

- Harmonize migration timelines
- Prevent regulatory arbitrage
- Share threat intelligence
- Align standards for cross-border interoperability

Existing forums like the G7, G20, and BIS are well placed to coordinate current mechanisms may not be fast or deep enough.

The Path Forward: Rebuilding Trust Before It Breaks

Quantum computing is not a distant cyber risk. It is a systemic transfo event. The foundations of digital trust — public-key cryptography — ar countdown clock.

The institutions and jurisdictions that act *now* will be the ones that:

- Preserve trust
- Maintain access to global networks
- Attract capital
- Lead standard-setting

Those that delay will face not just data breaches, but exclusion: from p systems, from counterparties, from trust itself.

This is the paradox of quantum: it is both the greatest systemic risk since 2008, and the greatest opportunity to rebuild financial infrastructure on secure, more agile foundations.

The question is not whether the system is at risk. It is whether we will time.

At-a-Glance: Quantum Risk vs Traditional Systemic

Final Note

The quantum revolution is both an existential risk and a generational opportunity. Those who prepare will not only secure resilience, they will redefine leadership in the financial system of the future. Those who do

learn — too late — that systemic risk in the quantum era is not about u
It is about *when*.

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